

NARA, Mali

WMO: 612330

Lat: 15.167N

Lon: 7.283W

Elev: 265

StdP: 98.18

Time Zone: 0.00 (E00)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.5	14.9	-10.5	1.6	29.0	-8.5	1.9	29.3	6.2	27.8	5.5	27.3	0.7	40	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	12.7	43.2	20.8	42.3	20.9	41.5	20.8	26.6	32.6	26.2	32.1	25.8	31.7	2.3	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.2	21.1	28.9	24.8	20.5	28.4	24.4	20.0	28.3	84.9	32.5	82.8	32.1	81.2	31.7	31.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.5	5.4	4.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	29.5	23.3	26.0	30.6	33.3	35.2	34.4	30.0	28.3	29.5	31.0	27.7	24.2	(6)	
(7)		DBStd	4.29	2.16	3.29	2.93	2.35	2.12	2.29	2.26	1.78	1.94	1.43	1.84	2.53	(7)	
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
(9)		HDD18.3	2	1	0	0	0	0	0	0	0	0	0	0	1	(9)	
(10)		CDD10.0	7113	413	449	638	701	781	732	621	567	587	651	532	440	(10)	
(11)		CDD18.3	4072	156	216	380	451	523	482	363	309	337	393	282	182	(11)	
(12)		CDH23.3	58186	1655	2678	5726	7539	8959	7907	4896	3487	4143	5523	3677	1998	(12)	
(13)		CDH26.7	36198	709	1486	3711	5280	6520	5543	2691	1581	2171	3383	2157	967	(13)	
(14)	Wind	WSAvg	1.8	1.9	2.2	2.3	2.2	2.0	2.2	1.8	1.4	1.3	1.4	1.8	(14)		
(15)	Precipitation	PrecAvg	393	0	0	1	3	11	40	114	140	72	15	1	0	(15)	
(16)		PrecMax	597	3	2	9	31	56	115	212	245	188	56	12	2	(16)	
(17)		PrecMin	199	0	0	0	0	0	1	35	37	14	0	0	0	(17)	
(18)		PrecStd	91	1	0	2	6	12	27	41	54	36	16	3	0	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.1	40.1	42.0	43.6	44.5	44.0	40.1	37.2	39.5	40.5	38.7	36.5	(19)	
MCWB			15.7	18.9	18.3	19.7	21.0	21.2	24.3	25.1	23.9	20.8	18.4	17.3	(20)		
2%		DB	33.4	37.8	40.7	42.6	43.4	42.7	38.0	35.2	37.5	39.5	37.7	35.0	(21)		
		MCWB	15.2	17.4	18.6	19.8	21.2	21.4	24.2	25.5	24.7	20.6	18.1	16.8	(22)		
5%		DB	32.1	36.2	39.6	41.7	42.6	41.7	36.5	34.0	36.1	38.7	36.8	33.7	(23)		
		MCWB	14.6	16.5	17.9	19.7	21.3	22.1	24.4	25.4	25.1	20.6	18.0	16.5	(24)		
10%		DB	30.7	34.4	38.5	40.7	41.6	40.5	35.3	33.0	34.8	37.9	35.7	32.0	(25)		
		MCWB	14.1	15.7	17.5	19.5	21.3	22.5	24.5	25.3	25.4	20.9	18.0	15.6	(26)		
(27)		Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.1	20.1	23.1	23.8	25.3	25.9	27.0	27.2	27.4	26.2	22.5	19.3	(27)
MCDWB				30.5	37.4	37.9	37.4	36.3	32.9	31.9	32.8	34.1	34.1	32.4	33.9	(28)	
2%	WB		16.3	18.2	20.8	22.7	24.4	25.1	26.1	26.4	26.7	25.1	21.0	17.9	(29)		
	MCDWB		29.5	35.7	34.7	38.0	36.0	34.6	32.0	31.9	32.9	32.7	31.3	33.2	(30)		
5%	WB		15.3	17.1	19.2	22.1	23.9	24.6	25.6	26.1	26.2	24.6	19.7	16.8	(31)		
	MCDWB		30.5	34.4	37.8	37.6	36.0	34.6	31.9	31.5	32.3	32.2	31.9	32.2	(32)		
10%	WB		14.5	16.1	18.1	21.3	23.5	24.1	25.3	25.7	25.8	24.0	18.5	16.0	(33)		
	MCDWB		29.9	32.9	36.9	37.1	36.0	34.5	31.8	30.9	31.8	31.9	33.0	31.1	(34)		
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.1	15.2	15.3	14.6	12.7	11.4	9.3	8.7	10.6	14.5	16.4	15.4	(35)	
MCDBR			17.3	17.0	15.7	14.8	13.8	12.5	10.7	10.0	12.3	15.7	17.7	17.4	(36)		
MCWBR			6.8	6.2	5.7	4.8	4.5	3.6	2.7	2.9	3.4	4.1	5.8	6.6	(37)		
5% WB		MCDBR	16.2	15.8	14.8	12.1	11.7	11.1	9.2	9.2	10.6	12.5	15.6	16.6	(38)		
		MCWBR	6.6	6.2	5.9	4.7	4.0	3.4	3.1	3.3	3.9	3.5	5.7	6.7	(39)		
		(40)	Clear-Sky Solar Irradiance	taub	0.506	0.532	0.643	0.657	0.699	0.720	0.651	0.593	0.581	0.572	0.494	0.473	(40)
taud	1.862	1.791		1.573	1.596	1.537	1.550	1.712	1.849	1.855	1.831	1.923	1.957	(41)			
Ebn at Noon	781	783		711	702	662	640	689	738	747	740	786	798	(42)			
Edh at Noon	196	220		280	274	286	280	239	210	208	208	183	174	(43)			
(44)	All-Sky Solar Radiation	RadAvg	5.28	5.92	6.57	6.89	6.57	6.24	5.99	5.92	6.08	5.90	5.46	5.13	(44)		
(45)		RadStd	0.33	0.32	0.31	0.27	0.34	0.20	0.21	0.18	0.25	0.15	0.21	0.21	(45)		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page